



black hat[®] EUROPE 2025

DECEMBER 8-11, 2025

EXCEL LONDON / UNITED KINGDOM

AI VS. AI

Building Resilient Enterprises in the Age of Autonomous Threats



DINIS CRUZ

GenAI Entrepreneur & Former CISO Akeia.ai /
The Cyber Boardroom / MyFeeds.ai

THE PERFECT STORM

Why This Moment Is Different

CONVERGING FORCES

- Weak Software Foundations
 - No liability, loose standards, chronic neglect of NFRs
- Business & Organizational Gaps
 - Tech as second-class concern, inadequate security funding
- Mature Attacker Ecosystem
 - \$10.5 trillion annually by 2025, commoditized attack tools
- Emergence of Autonomous AI Agents
 - Creative, persistent, finds alternate paths around restrictions

WE'VE BEEN GETTING AWAY WITH IT

- AWS: Timing bug wiped out a global database
- Azure: Config inconsistency silently propagated, then 'detonated' worldwide
- Cloudflare: Routine config change crashed bot defense, halted traffic
- CrowdStrike: 8.5 million Windows systems bricked by a faulty update

THE NEW INSIDERS

Autonomous Agents as Threat Actors

THREE TYPES OF AI-DRIVEN THREATS

- Malicious AI Agents
 - Fully autonomous malware that learns and adapts
- Compromised Benign Agents
 - Your AI bots hijacked via prompt injection
 - Operating with legitimate access, working for adversary
- Unintentionally Harmful Agents
 - The paperclip maximizer scenario
 - Dangerously efficient in ways we didn't anticipate

HUMAN INSIDERS VS. AI AGENTS

Human Insiders

- Natural limitations on skill/time
- Rarely had maximal intent
- Fatigue, fear of detection
- Single-threaded attacks

AI Agents

- Tireless, infinite patience
- Machine-speed adaptation
- Finds alternate paths when blocked
- No moral compass unless constrained

WHY ENTERPRISES AREN'T READY

Fragile by Design

BRITTLE ARCHITECTURE

- Technical Debt Crisis
 - ~40% of IT asset value is technical debt (McKinsey)
 - Systems running 'hot' with minimal safety margins
- Hidden Interdependencies
 - Internet = circular dependency machine
 - Identity, DNS, cloud, CDN all rely on each other
- Insufficient Isolation
 - Few hard boundaries inside to contain behavior
 - Not architected for machine-speed internal actors

THE MEASUREMENT PROBLEM

- Only ~54% have documented disaster recovery plans
- Many never test if their backups actually work
- Risk registers live in spreadsheets, disconnected from reality
- Near-misses treated as one-off glitches, not systemic warnings
- No version control for data and configurations

THE DEFENDER'S EDGE

AI as Force Multiplier, Not Black Box

SECURITY'S ASYMMETRIC ADVANTAGE

- Unique Enterprise-Wide Vantage Point
 - Access to every system, every data flow
 - One of few functions focused on business survival
- You Know What 'Good' Looks Like
 - Attackers eventually must do something outside the norm
 - You define the baseline - they have to deviate from it
- AI Finally Enables Modeling This
 - Digital twin of the enterprise
 - Process visibility and provenance at scale

AI FOR UNDERSTANDING, NOT BLOCKING

- Digital Twins & Dynamic Mapping
 - Living model of systems, data flows, dependencies
 - Highlight when anything strays outside known-good pattern
- Process Visibility and Provenance
 - Which agent did what, with which data, when
 - Convert logs into richly linked, queryable datasets
- Dynamic Threat Modeling
 - AI-simulated red team exercises - continuously
 - Discover attack paths before adversaries do

RADICAL SIMPLIFICATION

The Answer Isn't More Complexity

ASSUME COMPROMISE, CONTAIN BLAST RADIUS

- Strong Segmentation Inside
 - Crown-jewel systems walled off with strict controls
 - One-way isolation for legacy components
- Graceful Degradation = Victory
 - Parts may fail, but they fail safely
 - No cascade into total enterprise failure
- Hardened Boundaries Around What Can't Be Fixed
 - API throttles for fragile older systems
 - Circuit breakers that halt abnormal transactions

FUND NFRS AT UNPRECEDENTED SCALE

- Comprehensive backup and restore drills - actually test them
- Chaos engineering beyond normal peaks
- Instrumentation everywhere for real-time visibility
- Version control for data, not just code
- Consolidate overly complex workflows - fewer moving parts

DON'T BECOME THE DEPARTMENT OF 'NO'

- The Horse Has Bolted
 - Employees will use AI regardless
 - Shadow-IT AI is worse than governed AI
- Security as Enabler
 - Provide vetted platforms and guidelines
 - Safe internal AI services so they don't run to ChatGPT
- Govern 'Vibe Coding'
 - Non-engineers creating apps via AI assistance is reality
 - Establish guardrails and maturity models, don't outlaw it

THE PATH FORWARD

From Reactive Firefighting to Proactive Containment

KEY TAKEAWAYS

- AI vs. AI isn't about black-box defenses
 - It's about using AI to finally understand how enterprises actually work
- The race isn't to deploy GenAI defenses faster
 - Every agent becomes part of your attack surface
- Resilience beats prevention
 - Assume compromise, model normal, catch deviation
- Security's edge: knowing what 'good' looks like
 - Neither attackers nor runaway agents have anywhere to hide

CALL TO ACTION

- Build your digital twin now - don't wait for a crisis
- Invest in visibility, resilience, and continuous learning
- Simplify critical paths - remove unnecessary dependencies
- Bridge the gap between risk registers and technical reality
- Position security as the enabler of safe AI adoption

QUESTIONS?

@DinisCruz | The Cyber Boardroom | Akeia.ai